SUPPLEMENTARY DOCUMENT

GLOSSARY

Terms	Explanation
TF-IDF	Term Frequency Inverse Document Frequency
Weighted TF-IDF	An extension of TF-IDF with additional weighting to improve representation of minority classes in imbalanced dataset
NLP	Natural Language Processing
Stemming	A text preprocessing technique to reduce word to its root form.
OvR (One vs Rest)	A strategy for multiclass classification where 'n' no. classifiers are trained for 'n' classes.
OvO (One vs One)	A strategy where classifier is trained for every pair of classes. [no. of classifiers: K(K-1)/2; K: no. of classes]
Kernel Trick	A technique used to transform data into a higher dimensional space to facilitate linear separation
Grid Search	A systematic method of testing multiple hyperparameters combinations to find the optimal model configuration.
Hyperplane	Decision boundary used by SVM to separate data points into different classes in a high dimensional space.
Dropout	A regularisation technique where random % of neurons are ignored during training to tackle overfitting.
Weight decay	A regularisation technique where large weights are penalised more.
Jaccard Score (Jaccard Accuracy)	Evaluates classification performance: Formula: size of intersection/size of union
F1 (Macro)	A performance metric that combines precision and recall Macro: averaged equally across all classes

ADDITIONAL INFORMATION

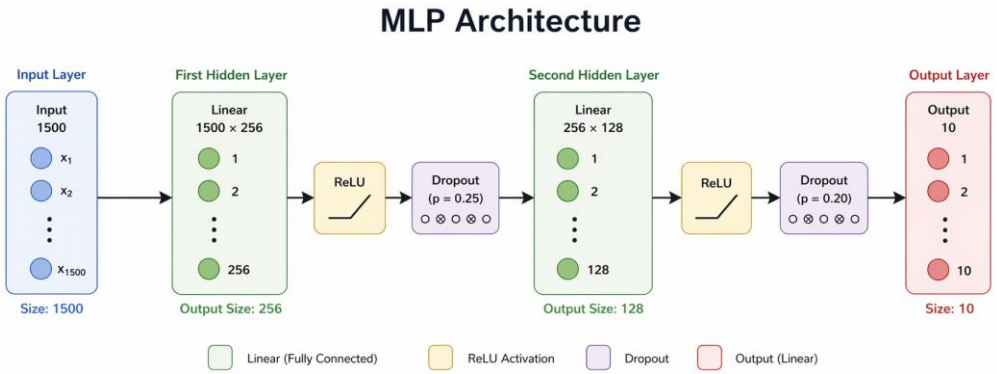


Fig.1: Implemented MLP architecture (AI generated diagram)

In this analysis, MLP was the first model to be tested on the Reuters dataset. The default settings for MLP were used with 256x128 architecture and basic traditional TF-IDF was implemented, and results were really disheartening (Fig 1). After that the tuned MLP was also implemented but results were still weak (Fig 2).

While training SVM and tuning for C (regularisation parameter), increasing C is continuously improving the training score of the model, as it was learning all the noise as well which motivated me to plot the training and validation for the SVM configurations (Fig 3(a)(b)(c)). Hence, we restricted with optimal hyperparameter values instead of best parameters with highest score.

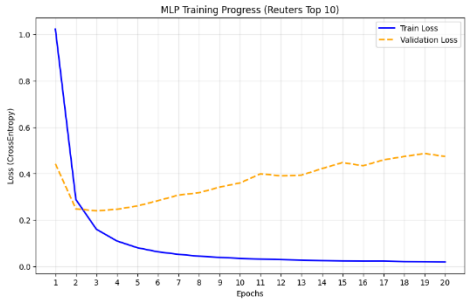


Fig.1: Default MLP with Traditional TF-IDF

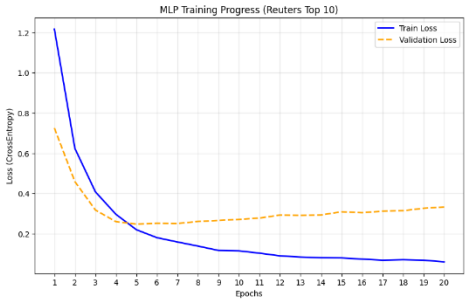


Fig.2: Tuned MLP with Traditional TF-IDF

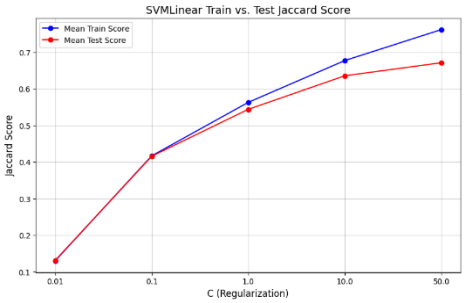


Fig.3(a): Regularisation tuning score for Linear SVM

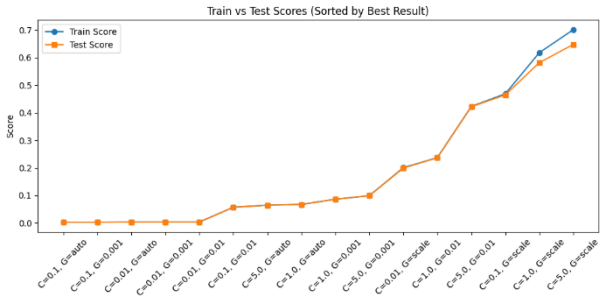


Fig.3(b): Hyperparameters tuning for RBF SVM

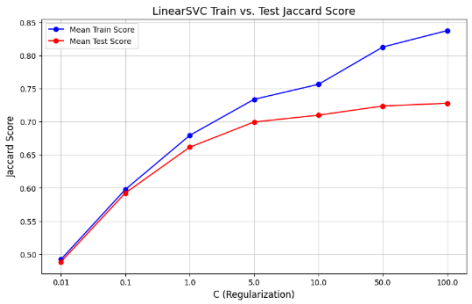


Fig.3(c): Regularisation tuning for LinearSVC